📘 **AI/ML Internship – Day 57 Notes & Tasks**

🟢 **Module 7: Pre-trained Word Embeddings (GloVe & FastText)**

📌 Part 1: Recap of Day 56

Yesterday we learned:

✅ Word Embeddings

✅ Word2Vec

✅ CBOW

✅ Skip-Gram

✅ Semantic Relationships

Today we will learn:

🔥 Pre-trained Word Embeddings

🔥 GloVe

🔥 FastText

These techniques are widely used in real-world NLP projects.

📌 Part 2: Why Do We Need Pre-trained Embeddings?

Suppose you want to build:

- ChatGPT-like chatbot
- Spam Detector
- News Classifier
- Resume Analyzer

Can you train Word2Vec from scratch?

✅ Yes.

But...

Imagine you have only:

1000 sentences

Is that enough?

❌ No.

Word2Vec performs best with millions of sentences.

📌 Part 3: What are Pre-trained Embeddings?

Definition

Pre-trained Word Embeddings are word vectors already trained on very large datasets.

Instead of training again, we simply download and use them.

Real-Life Example

Imagine learning English.

Option 1:

Read every dictionary yourself.

Takes years.

Option 2:

Buy an already published dictionary.

Much faster.

Pre-trained embeddings are like the second option.

📌 Part 4: Popular Pre-trained Embeddings

The three most popular are:

Model	Developed By	Purpose
Word2Vec	Google	Word Embeddings
GloVe	Stanford University	Global Word Representation
FastText	Meta (formerly Facebook AI)	Handles rare and unknown words

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📌 Part 5: What is GloVe?

Full Form

Global Vectors for Word Representation

Definition

GloVe is a pre-trained word embedding model that learns word meanings using global word co-occurrence statistics.

Simple Meaning

Instead of looking only at nearby words, GloVe learns from how often words appear together across the entire dataset.

📌 Part 6: Example of GloVe

Dataset:

The king rules the kingdom.

The queen rules the kingdom.

The prince lives in the palace.

GloVe learns:

King ↔ Queen

King ↔ Palace

Queen ↔ Palace

Because these words frequently appear together.

📌 Part 7: How GloVe Works

Workflow:

Large Dataset
↓
Count Word Co-occurrences
↓

Build Co-occurrence Matrix
↓
Generate Word Vectors
↓
Use in NLP Models

📌 Part 8: Advantages of GloVe

✅ Understands semantic meaning

✅ Captures global relationships

✅ High accuracy

✅ Widely used

📌 Part 9: Limitation of GloVe

Suppose word:

ChatGPT

doesn't exist in the pre-trained vocabulary.

Result:

❌ No vector available.

This is called:

Out of Vocabulary (OOV)

📌 Part 10: What is FastText?

FastText is another pre-trained embedding model developed by Meta AI.

Unlike Word2Vec and GloVe,

FastText breaks words into smaller pieces called **character n-grams**.

📌 Part 11: Character n-grams

Example:

Word:

Playing

FastText creates pieces like:

Pla

Play

Lay

ayi

ying

These small pieces help FastText understand unknown words.

📌 Part 12: Why is FastText Better?

Suppose:

Training data contains:

play

Testing word:

playing

Word2Vec:

❌ May not know "playing" if it wasn't seen.

FastText:

✅ Understands because it recognizes character patterns.

📌 Part 13: Another Example

Known word:

computer

Unknown word:

computers

FastText recognizes:

computer

inside

computers

So it can generate a meaningful vector.

📌 Part 14: GloVe vs FastText

GloVe	FastText
Uses whole words	Uses character n-grams
Faster inference	Better for rare words
Cannot handle unknown words well	Handles unknown words effectively
Global statistics	Subword information

📌 Part 15: Real-World Example

Sentence:

Artificial Intelligence is amazing.

If "Artificial" is unknown:

GloVe:

❌ May fail.

FastText:

✅ Uses character pieces.

Produces a meaningful vector.

📌 Part 16: Using GloVe in Python

Install:

pip install gensim

Example:

```
import gensim.downloader as api

model = api.load("glove-wiki-gigaword-50")

print(model["king"])
```

Output:

A vector with 50 numerical values.

📌 Part 17: Finding Similar Words

```
print(model.most_similar("king"))
```

Example Output:

queen

prince

monarch

royal

📌 Part 18: Using FastText

Example:

```
import gensim.downloader as api

model = api.load("fasttext-wiki-news-subwords-300")
```

Find similar words:

```
print(model.most_similar("computer"))
```

📌 Part 19: Applications

Search Engines

Finding related search terms.

Chatbots

Understanding user intent.

Machine Translation

Translating between languages.

Text Classification

News classification.

Sentiment Analysis

Understanding review meanings.

Question Answering

Finding semantically related answers.

📌 Part 20: Comparison

Word2Vec	GloVe	FastText
Learns local context	Learns global context	Learns character-level information
Needs training	Pre-trained available	Pre-trained available
Weak for unknown words	Weak for unknown words	Excellent for unknown words

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📌 Part 21: Theory Questions

What are Pre-trained Word Embeddings?

Word vectors already trained on large datasets and ready to use.

What is GloVe?

A pre-trained embedding model based on global word co-occurrence.

What is FastText?

A word embedding model that uses character n-grams.

What is OOV?

Out of Vocabulary – words not present in the trained vocabulary.

Why is FastText better for unknown words?

Because it learns from character-level information.

📌 Part 22: Interview Questions

Difference between Word2Vec and GloVe?

Word2Vec learns from local context, while GloVe learns from global co-occurrence statistics.

Difference between GloVe and FastText?

GloVe represents complete words, whereas FastText represents words using character n-grams.

What is an Out-of-Vocabulary (OOV) word?

A word that is not present in the model's vocabulary.

Why are pre-trained embeddings useful?

They save training time and provide high-quality word representations.

Which embedding model is better for misspelled or rare words?

FastText.

🎯 Tasks for Day 57

🧠 Theory

1. What are Pre-trained Word Embeddings?
2. Explain GloVe.
3. Explain FastText.
4. What is OOV?
5. Compare Word2Vec, GloVe, and FastText.

💻 Practical Task 1

Install Gensim:

pip install gensim

💻 Practical Task 2

Load the GloVe model.

💻 Practical Task 3

Print the vector for:

king

💻 Practical Task 4

Find the top 5 similar words for:

computer

💻 Practical Task 5

Research one real-world application of FastText and explain why it is suitable.